

Remote Controlled CORONA PROTECTION BELL

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The World Health Organization (WHO) recommends that everyone should rub their hands with soap for at least 20 seconds before washing with water to prevent transmission of the novel Corona virus, which is again on rise, in addition to social distancing. The 20-second period can be checked with wristwatches and the stopwatches on mobile phones, but these are not convenient to use while washing hands. This proposed circuit rings a buzzer for 20 seconds and switches on an LED for audio as well as visual indication (for those with hearing issues). The author's

prototype is shown in Fig. 1.

The circuit for this alarm, shown in Fig. 2, comprises a 9V battery, 5V voltage regulator LM7805 (IC1), timer NE555 (IC2), relay driver ULN2003 (IC3), 5V SPDT relay (RL1), and a few other components. Resistor R2 and capacitor C2 decide the time of monostable, which is around 20 seconds here. This output time period of IC2 at its pin 3 (monostable multivibrator built around NE555) can be calculated using following relationship:

$$T = 1.1RC$$

where $R = R2 = 18k$ and $C = C2 = 1000\mu F$.

So, time (T) is equal ($= 1.1 \times 18k \times 1000\mu F$) to around 20 seconds.

The infra-red receiver TSOP1738 generates an active-low output. When the remote is targeted at

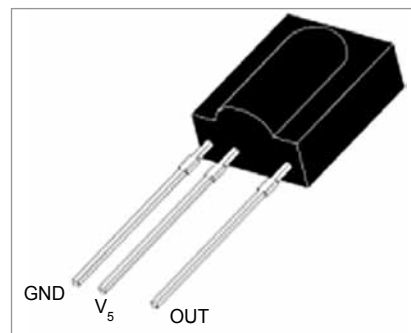


Fig. 3: Image of TSOP1738

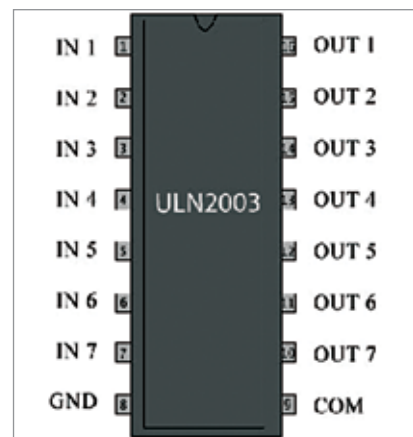


Fig. 4: Pin diagram of ULN2003

TSOP1738 and any key is pressed, input pin 1 of ULN2003 relay driver gets activated and drives the relay.

As a result, IC2 produces 20-second output for switching on LED1 as well as the buzzer.

A series of TSOP17XX receivers is available with divergent carrier frequencies for use in television sets, DVD players, etc, to

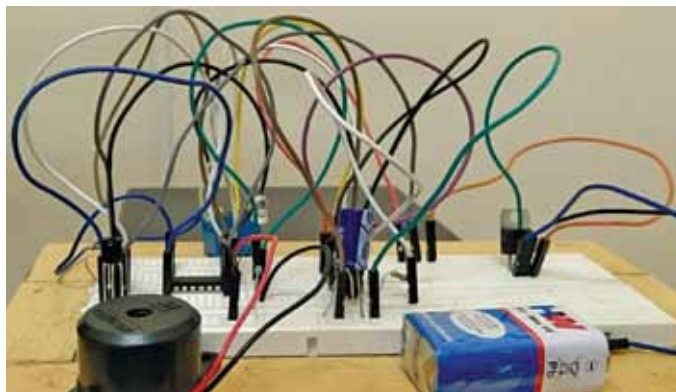


Fig. 1: Author's prototype on a breadboard

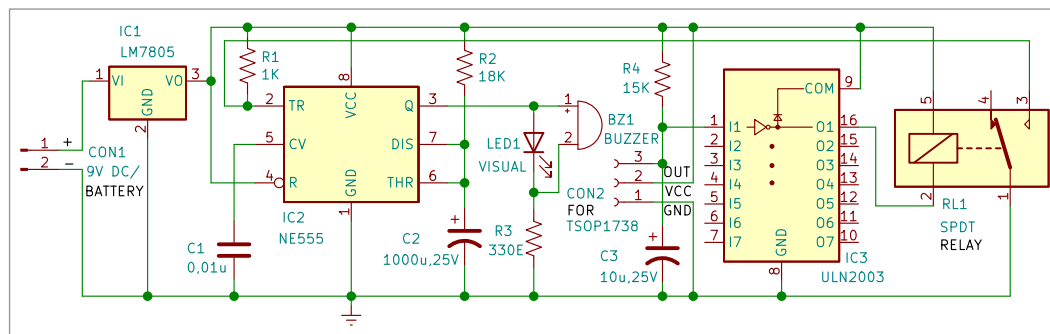


Fig. 2: Circuit diagram of the alarm